

Tugas Kalkulus

1.) $y = \sin(2x-1)$

$y = \sin u \rightarrow y' = \cos u$

$u = 2x-1 \rightarrow u' = 2$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$= \cos u \cdot 2$

$= \cos 2x-1 \cdot 2$

$= 2 \cos(2x-1)$

$y = 2 \cos \rightarrow -2 \sin$

$u = 2x-1 \rightarrow 2$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$= -2 \sin \cdot 2$

$= -4 \sin 2x-1$

2.) $y = (2x-3)^3$

$8(2x-3)^3$

$y = u^3 \rightarrow 3u^2$

$y = 8(u)^3 \rightarrow 24(u)^2$

$u = (2x-3) \rightarrow 2$

$u = 2x-3 \rightarrow 2$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$= 3u^2 \cdot 2$

$= 24u^2 \cdot 2$

$= 4(2x-3)^2 \cdot 2$

$= 24(2x-3)^2 \cdot 2$

$= 8(2x-3)^2 \text{ (turunan Pertama)}$

$= 48(2x-3)^2$

$$3.) y = \frac{x}{(x+1)} \quad u = x - 0 \quad 1$$
$$v = (x+1) - 0 \quad 1$$

$$\frac{u}{v} = \frac{u'v - v'u}{v^2}$$

$$= \frac{1(x+1) - 1(x)}{(x+1)^2}$$

$$= \frac{x+1-1}{(x+1)^2}$$

$$= \frac{x-x+1}{(x+1)^2}$$

$$= \frac{1}{(x+1)^2} = (x+1)^{-2} \text{ turunan pertama}$$

$$(x+1)^{-2}$$

$$y = u^{-2} \rightarrow -2u^{-3}$$

$$u = x+1 \rightarrow 1$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= -2u^{-3} \cdot 1$$

$$= -2(x+1)^{-3} = \frac{-2}{(x+1)^3}$$

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$$4.) 4x^2y - 3y = x^3 - 1$$

$$(8xy + 4x^2y') - 3y' = 3x$$

$$\cancel{8x} y' (4x^2 - 3) = 3x - \cancel{8x} y$$

$$y' = \frac{3x - 8xy}{4x^2 - 3}$$